BSE4104 – EMERGING TRENDS IN SOFTWARE ENGINEERING

WEEK 2 PROGRESS REPORT

Project Title	CampusConnect: A University Student-Support Agent for CoCIS Freshers.
Week	Foundation-Model Engineering and Prompting
Group	Group A Evening
Team Members	
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1. Tasks Completed

- Selected Gemini 3.8 Flash as the foundation model for CampusConnect, considering its capabilities, accessibility and API usage costs.
- Integrated the Gemini API using the google-genai Python SDK to establish the initial model interaction.
- Prepared the Prompt Specification and Version History document, defining the assistant's role, task, context, constraints, output format and failure behaviour.
- Defined prompt instructions to guide the assistant in providing relevant and concise responses to CoCIS-related student questions while avoiding unsupported information.
- Prepared the Model Selection Note to document the selected model and the considerations behind the choice.
- Created and completed the 10-case prompt evaluation table, comparing expected and actual model behaviour.
- Tested the model's responses and reviewed prompt iterations to assess compliance with the prompt instructions and identify areas for improvement.

2. Tasks in Progress

- Preparing the Corpus Register to document the selected documents for the CampusConnect knowledge base, including their titles, sources and relevance to the project.
- Finalising the Corpus Provenance documentation to record the origins of the selected knowledge sources and provide information about their suitability for use in CampusConnect.

3. Next Week's Tasks

- Complete the Corpus Register and Corpus Provenance documentation and organise the selected documents into a controlled knowledge corpus.
- Design and implement a Retrieval-Augmented Generation (RAG) pipeline for CampusConnect.
- Implement document processing, chunking, indexing and retrieval to identify relevant information from the knowledge corpus.
- Integrate retrieved information into the model's responses to improve factual grounding.
- Create 15 RAG evaluation questions covering answerable, partially answerable and unanswerable cases.
- Evaluate retrieval and response grounding, identifying at least three retrieval or grounding failures.
- Prepare the RAG architecture diagram and Week 3 progress report.

4. Problems / Challenges Faced

- Selecting a foundation model that was accessible for development while considering API costs and usage limitations.
- Defining prompt instructions that guide the assistant to provide relevant student-support information without inventing university-specific details.
- Establishing clear failure behaviour for questions that cannot be answered using the available information.
- Recognising the need to evaluate actual model responses rather than relying only on the successful execution of the API call.
- Identifying the need for reliable, traceable knowledge sources to support the assistant's responses as the project progresses towards RAG implementation.